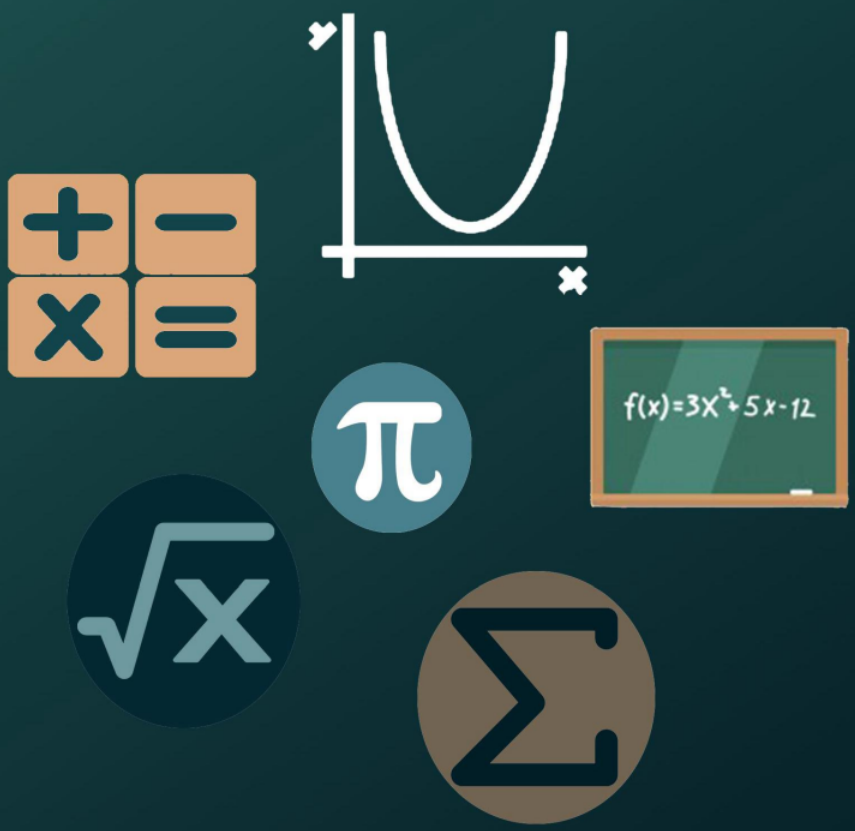
MATHEMATICS

Application of Derivative



SHEET

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Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 7
5.	Exercise 2	Page - 12
6.	Exercise 2A	Page - 20
7.	Answer Key	Page - 26

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Nature of a Function at a Point				
2	Increasing and Decreasing Nature of a Function over an Interval	1-6	6-8	1-4	1-5, 31, 36
3	Problems based on Monotonicity of Function	7-9	1	5-12	29
4	Greatest and Least Value of a Function	10		13, 14	
5	Establishing Inequalities	11		15, 16, 35	6, 8, 48
6	Rolle's Theorem	12, 13	30 - 33	17-24	7, 34
7	Lagrange's Mean Value Theorem	14-16	2 - 3, 20, 34 - 37	25-29	
8	Introduction to Maxima & Minima				
9	Derivative Test For Ascertaining Maxima /Minima	17-22	25, 26, 38, 39	30-34, 36, 37, 39, 41	9-11, 28, 38-40
10	Special Point on A Given Curve	23, 24	21, 40 - 43	38, 40, 42-50	12-14, 25-27
11	Optimization Problems	25-29	4, 9 - 11, 27, 44	51-56	41, 42
12	Maximum & Minimum Distance of a Fixed Point From a Given Curve	30	22		44
13	Roots of a Cubic Equation	31	12 - 13	57, 58	32, 33, 35, 37, 45
14	Rate Measure	32-36	45, 46	59-62	15
15	Approximation	37		63	
16	Tangent and Normal at a Point	38	5, 28, 47 - 51	64-74	22-24
17	Tangent and Normal in Parameric Form	55-58	52, 53		17
18	Equation of Tangent and Normal in Some Special Cases	59	23, 29	75-78	30, 46, 47
19	Angle Between Two Curves	60-63	14 - 19, 30, 54		
20	Length of Tangent, Normal, Subtangent and Subnormal	64-66	55, 56		16
21	After Completing Chapter		24		19-21, 43, 49

Note

All questions marked with * sign are to be attempted after completion of Integration.

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Monotocity

Increasing and Decreasing Nature of a Function over an Interval

Q1. A function is matched below against an interval where it is supposed to be increasing. Which of the following pairs is incorrectly matched ?

Interval	Function
(A) $\left(-\infty, \frac{1}{3}\right)$	$3x^2 - 2x + 1$
(B) $(-\infty, -4)$	$x^3 + 6x^2 + 6$
(C) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(D) $(2, \infty)$	$2x^3 - 3x^2 - 12x + 6$

Q2. Given function $f(x) = \left(\frac{e^{2x} - 1}{e^{2x} + 1}\right)$ is

(A) Increasing (B) Decreasing
(C) Even (D) None of these

Q3. The function $f(x) = x - \log(1 + x)$, $x > -1$ is increasing in the interval

(A) $(0, \infty)$ (B) $(-1, 0)$
(C) $(-\infty, 0)$ (D) None of these

Q4. The function x^x is increasing, when

(A) $x > \frac{1}{e}$ (B) $x < \frac{1}{e}$
(C) $x < 0$ (D) For all real x

Q5. $2x^3 + 18x^2 - 96x + 45 = 0$ is an increasing function when

(A) $x < -8, x > 2$ (B) $x < -2, x > 8$
(C) $x \le -2, x \ge 8$ (D) $-2 < x < 0$

***Q6.** Let $f(x) = \int e^x (x-1)(x-2)\,dx$.

Then f decreases in the interval :

(A) $(-\infty, -2)$ (B) $(-2, -1)$
(C) $(1, 2)$ (D) $(2, \infty)$

Problems based on Monotonicity of Function

Q7. Function $f(x) = \frac{\lambda \sin x + 6\cos x}{2\sin x + 3\cos x}$ is monotonic increasing, if

(A) $\lambda > 1$ (B) $\lambda < 1$
(C) $\lambda < 4$ (D) $\lambda > 4$

Q8. The values of 'a' for which the function $(a+2)x^3 - 3ax^2 + 9ax - 1$ decreases monotonically throughout for all real x, are

(A) $a \le -2$ (B) $a \ge -2$
(C) $-3 \le a < 0$ (D) $-\infty < a \le -3$

Q9. $f(x) = \frac{\ell n(1+x)}{x}, x \ne 0$ is -

(A) increasing for $x > 0$
(B) decreasing for $x > 0$ only
(C) increasing for all x in its domain
(D) decreasing for all x in its domain

Greatest and Least Value of a Function

Q10. Let $f(x) = 2x^2 + \frac{2}{x^2}, \quad 0 < |x| \le 2$

$= 1, \quad x = 0$

Then f(x) has

(A) least value 4 but no greater value
(B) greatest value but no least value
(C) neither least nor greatest value
(D) least value 1 but no greatest value

Establishing Inequalities

Q11. For all $x \in (0,1)$

(A) $e^x < 1+x$ (B) $\log_e(1+x) < x$

(C) $\sin x > x$ (D) $\log_e x > x$

Rolle's Theorem

Q12. Rolle's theorem is not applicable to the function $f(x) = |x|$ defined on $[-1, 1]$ because :

(A) f is not continuous on $[-1, 1]$

(B) f is not differentiable on $(-1,1)$

(C) $f(-1) \neq f(1)$

(D) $f(-1) = f(1) \neq 0$

Q13. If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval :

(A) (2, 3) (B) (1, 2)

(C) (0, 1) (D) (1, 3)

Lagrange's Mean Value Theorem

***Q14.** Let f be differentiable for all x. If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$ then-

(A) $f(6) \geq 8$ (B) $f(6) < 8$

(C) $f(6) < 5$ (D) $f(6) = 5$

Q15. A value of C for which the conclusion of Mean values theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is-

(A) $2\log_3 e$ (B) $\frac{1}{2}\log_e 3$

(C) $\log_3 e$ (D) $\log_e 3$

Q16. P (2, 2) and $Q\left(\frac{1}{2}, -1\right)$ are two points on the parabola $y^2 = 2x$. The coordinates of the point R on the parabola, where the tangent to the curve is parallel to the chord PQ, is

(A) $\left(\frac{5}{4}, \sqrt{\frac{5}{2}}\right)$ (B) $(2, -1)$

(C) $\left(\frac{1}{8}, \frac{1}{2}\right)$ (D) None of these

Derivative Test For Ascertaining Maxima-Minima

Q17. The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minimum at-

(A) $x = -2$ (B) $x = 0$

(C) $x = 1$ (D) $x = 2$

Q18. The value of the function $(x-1)(x-2)^2$ at its maxima is

(A) 1 (B) 2

(C) 0 (D) $\frac{4}{27}$

Q19. The function $x^5 - 5x^4 + 5x^3 - 10$ has a maximum, when x =

(A) 3 (B) 2 (C) 1 (D) 0

Q20. The function $x^2 \ell nx$ in the interval (1, e) has

(A) A point of maximum

(B) A point of minimum

(C) Points of maximum as well as of minimum

(D) Neither a point of maximum nor minimum

Q21. Local maximum value of the function $\frac{\ell nx}{x}$ is-

(A) e (B) 1 (C) $\frac{1}{e}$ (D) 2e

Q22. The maximum value of $x^4 e^{-x^2}$ is

(A) e^2 (B) e^{-2}

(C) $12e^{-2}$ (D) $4e^{-2}$

Special Point on A Given Curve

Q23. x^x has a stationary point at

(A) $x = e$ (B) $x = \frac{1}{e}$

(C) $x = 1$ (D) $x = \sqrt{e}$

Q24. For the function $f(x) = 2x^2 - \ln|x|$

(A) set of critical points is $\{-1/2, 0, 1/2\}$

(B) f(x) is increasing in $(-\infty, -1/2] \cup (0, 1/2]$

(C) f(x) is decreasing in $[-1/2, 0) \cup [1/2, \infty)$

(D) None of these

Optimization Problems

Q25. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x. The maximum area enclosed by the park is-

(A) $\sqrt{\frac{x^3}{8}}$ (B) $\frac{1}{2}x^2$

(C) πx^2 (D) $\frac{3}{2}x^2$

Q26. If $x + y = 16$ and $x^2 + y^2$ is minimum, then the values of x and y are

(A) 3, 13 (B) 4, 12

(C) 6, 10 (D) 8, 8

Q27. The sum of two non-zero numbers is 4. The minimum value of the sum of their reciprocals is -

(A) $\frac{3}{4}$ (B) $\frac{6}{5}$

(C) 1 (D) None of these

Q28. If from a wire of length 36 metre a rectangle of greatest area is made, then its two adjacent sides in metre are -

(A) 6, 12 (B) 9, 9

(C) 10, 8 (D) 13, 5

Q29. A cone of maximum volume is inscribed in a given sphere, then ratio of the height of the cone to diameter of the sphere is -

(A) 2/3 (B) 3/4

(C) 1/3 (D) 1/4

Maximum & Minimum Distance of a Fixed Point From a Given Curve

Q30. The shortest distance of the point (0, 0) from the curve $y = \frac{1}{2}\left(e^x + e^{-x}\right)$ is

(A) 2 (B) 1

(C) 3 (D) None of these

Roots of a Cubic Equation

Q31. Suppose the cubic $x^3 - px + q$ has three real roots where $p > 0$ and $q > 0$. Then which of the following holds ?

(A) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$.

(B) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$.

(C) The cubic has minima at both $-\sqrt{\frac{p}{3}}$ & $\sqrt{\frac{p}{3}}$.

(D) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ & $-\sqrt{\frac{p}{3}}$.

Tangent - Normal and Rate Measure

Rate Measure

Q32. If a particle is moving such that the velocity acquired is proportional to the square root of the distance covered, then its acceleration is-

(A) A constant (B) $\propto s^2$

(C) $\propto \frac{1}{s^2}$ (D) $\propto \frac{1}{s}$

Q33. For what values of x is the rate of increase of $x^3 - 5x^2 + 5x + 8$ is twice the rate of increase of x ?

(A) $-3, -\frac{1}{3}$ (B) $-3, \frac{1}{3}$

(C) $3, -\frac{1}{3}$ (D) $3, \frac{1}{3}$

Q34. A particle moves in a straight line in such a way that its velocity at any point is given by $v^2 = 2 - 3x$, where x is measured from a fixed point. The acceleration is -

(A) Uniform (B) Zero

(C) Non-uniform (D) Indeterminate

Q35. If a spherical balloon has a variable diameter $3x + \frac{9}{2}$, then the change of its volume with respect to x is

(A) $27\pi(2x+3)^2$ (B) $\frac{27\pi}{16}(2x+3)^2$

(C) $\frac{27\pi}{8}(2x+3)^2$ (D) None of these

Q36. If by dropping a stone in a quiet lake a wave moves in circle at a speed of 3.5 *cm/sec*, then the rate of increase of the enclosed circular region when the radius of the circular wave is 10 *cm*, is $\left(\pi = \frac{22}{7}\right)$

(A) 220 sq. cm/sec

(B) 110 sq. cm/sec

(C) 35 sq. cm/sec

(D) 350 sq. cm/sec

Approximation

Q37. The value of $(127)^{1/3}$ to four decimal places is -

(A) 5.0267 (B) 5.4267

(C) 5.5267 (D) 5.001

Tangent and Normal at a Point

Q38. The equation of tangent to the curve $y^2 = 6x$ at $(2, -3)$

(A) $x + y - 1 = 0$ (B) $x + y + 1 = 0$

(C) $x - y + 1 = 0$ (D) $x + y + 2 = 0$

Q39. The equation of the tangent to the curve $x^2(x - y) + a^2(x + y) = 0$ at origin is-

(A) $x + y + 1 = 0$ (B) $x + y + 2 = 0$

(C) $x + y = 0$ (D) $2x - y = 0$

Q40. The point of contact on the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ where-

(i) tangent line is parallel to x-axis

(ii) tangent line is parallel to y-axis

(ii) tangent line makes equal angles with both the axis

(A) (a, 0), (0, a), (a/4, a/4)

(B) (0, a), (a, 0), (a/4, a/2)

(C) (0, a), (a/4, a/4), (a, a)

(D) (0, 0), (a, 0), (0, a)

Q41. The length of intercepts on coordinate axes made by tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point $P(x_1, y_1)$ are-

(A) $\sqrt{ax_1}, \sqrt{ay_1}$ (B) $\sqrt{a}, \sqrt{a}$

(C) $\sqrt{x_1}, \sqrt{y_1}$ (D) None of these

Q42. The length of intercepts on coordinate axes made by tangent to the curve $y = 2x^2 + 3x - 2$ at the point (1, 3) are-

(A) 4, –4/7 (B) –4/7, 4

(C) 4/7, –4 (D) 4/7, 4

Q43. If the normal to the curve, y = f(x) at the point (3, 4) makes an angle $\frac{3\pi}{4}$ with the positive x-axis.Then f'(3) -

(A) –1 (B) $-\frac{3}{4}$

(C) $\frac{4}{3}$ (D) 1

Q44. The equation of the normal to the curve $y = \sin\frac{\pi x}{2}$ at (1, 1) is

(A) $y = 1$

(B) $x = 1$

(C) $y = x$

(D) $y - 1 = \frac{-2}{\pi}(x - 1)$

Q45. The equation of tangent to the curve $y = 2\cos x$ at $x = \frac{\pi}{4}$ is

(A) $y - \sqrt{2} = 2\sqrt{2}\left(x - \frac{\pi}{4}\right)$

(B) $y + \sqrt{2} = \sqrt{2}\left(x + \frac{\pi}{4}\right)$

(C) $y - \sqrt{2} = -\sqrt{2}\left(x - \frac{\pi}{4}\right)$

(D) $y - \sqrt{2} = \sqrt{2}\left(x - \frac{\pi}{4}\right)$

Q46. The equation of the tangent to the curve $(1 + x^2)y = 2 - x$, where it crosses the x-axis, is

(A) $x + 5y = 2$

(B) $x - 5y = 2$

(C) $5x - y = 2$

(D) $5x + y - 2 = 0$

Q47. The tangent to the curve $y = ax^2 + bx$ at $(2, -8)$ is parallel to x-axis. Then

(A) $a = 2, b = -2$ (B) $a = 2, b = -4$

(C) $a = 2\ b = -8$ (D) $a = 4, b = -4$

Q48. At what point on the curve $x^3 - 8a^2y = 0$, the slope of the normal is $\frac{-2}{3}$

(A) (a, a) (B) $(2a, -a)$

(C) $(2a, a)$ (D) None of these

Q49. The equation of tangent to the curve $\frac{x^2}{3} - \frac{y^2}{2} = 1$ which parallel to y = x, is

(A) $y = x \pm 1$ (B) $y = x - 1/2$

(C) $y = x + 1/2$ (D) $y = 1 - x$

Q50. The points on the curve $y^2 = 4a\left(x + a\sin\frac{x}{a}\right)$ at which the tangent is parallel to x-axis, lie on -

(A) A straight line (B) A parabola

(C) A circle (D) An ellipse

Q51. The curve $y - e^{xy} + x = 0$ has a vertical tangent at the point -

(A) (1, 1) (B) At no point

(C) (0, 1) (D) (1, 0)

Q52. The tangent to the curve $3xy^2 - 2x^2y = 1$ at (1, 1) meets the curve again at :

(A) $\left(-\frac{16}{5}, -\frac{1}{20}\right)$ (B) $\left(\frac{16}{5}, -\frac{1}{20}\right)$

(C) $\left(-\frac{16}{5}, \frac{1}{20}\right)$ (D) None of these

Q53. At which point the line $\frac{x}{a}+\frac{y}{b}=1$, touches the curve $y = be^{-x/a}$

(A) (0, 0) (B) (0,a)
(C) (0, b) (D) (b, 0)

***Q54.** The slope of the tangent to the curve $y=\int_0^x \frac{dt}{1+t^3}$ at the point where x = 2 is -

(A) $\frac{1}{9}$ (B) 9

(C) $\frac{1}{3}$ (D) None of these

Tangent and Normal in Parameric Form

Q55. For the curve $x = t^2 - 1$, $y = t^2 - t$ the tangent line is perpendicular to x-axis, where-

(A) $t = 0$ (B) $t = \infty$

(C) $t = 1/\sqrt{3}$ (D) $t = -1/\sqrt{3}$

Q56. The normal to the curve $x = a(1 + \cos\theta)$, $y = a\sin\theta$ at 'θ' always passes through the fixed point-

(A) (a, 0) (B) (0, a)
(C) (0, 0) (D) (a, a)

Q57. The slope of tangent to the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ at the point (2, –1) is

(A) $\frac{22}{7}$ (B) $\frac{6}{7}$

(C) – 6 (D) None of these

Q58. The equation of the tangent to the curve $x = 2\cos^3\theta$ and $y = 3\sin^3\theta$ at the point $\theta = \pi/4$ is-

(A) $2x + 3y = 3\sqrt{2}$

(B) $2x - 3y = 3\sqrt{2}$

(C) $3x + 2y = 3\sqrt{2}$

(D) $3x - 2y = 3\sqrt{2}$

Equation of Tangent and Normal in Some Special Cases

Q59. Co-ordinates of a point on the curve $y = x\log x$ at which the normal is parallel to the line $2x - 2y = 3$ are

(A) (0,0) (B) (e, e)

(C) $(e^2, 2e^2)$ (D) $(e^{-2}, -2e^{-2})$

Angle Between Two Curves

Q60. The angle of intersection between the curves $y = x$ and $y^2 = 4x$ at (4, 4)

(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) $\tan^{-1}\left(\frac{1}{3}\right)$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Q61. The condition that the curves $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $xy = c^2$ mutually intersect orthogonally is-

(A) $a^2 + b^2 = 0$ (B) $a^2 = b^2$

(C) $a^2b^2 = 0$ (D) $a^2 = \frac{1}{-b^2}$

Q62. Angle between the tangents to the curve $y = x^2 - 5x + 6$ at the points (2, 0) and (3, 0) is-

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Q63. If the curve $y = a^x$ and $y = b^x$ intersect at angle α then, $\tan\alpha =$

(A) $\left|\frac{a-b}{1+ab}\right|$ (B) $\left|\frac{\ell n\, a - \ell n\, b}{1+\ell n\, a\, \ell n\, b}\right|$

(C) $\left|\frac{a+b}{1-ab}\right|$ (D) $\left|\frac{\ell n\, a + \ell n\, b}{1-\ell n\, a\, \ell n\, b}\right|$

Length of Tangent, Normal, Subtangent and Subnormal

Q64. The length of normal to the curve $x = a(\theta + \sin\theta),\ y = a(1 - \cos\theta)$ at the point $\theta = \pi/2$ is -

(A) $2a$ (B) $a/2$

(C) $\sqrt{2}\,a$ (D) $a/\sqrt{2}$

Q65. The length of the subtangent to the curve $x^2 + xy + y^2 = 7$ at $(1, -3)$ is -

(A) 3 (B) 5 (C) 15 (D) 3/5

Q66. For the curve $by^2 = (x+a)^3$ the square of subtangent is proportional to

(A) $(\text{Subnormal})^{1/2}$ (B) Subnormal

(C) $(\text{Subnormal})^{3/2}$ (D) None of these

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The funtion $f(x) = \sqrt{3}\ \sin x - \cos x$ will increases monotonically in the interval (s)

(A) $0 < x < \frac{\pi}{2}$ (B) $-\frac{\pi}{3} < x < \frac{2\pi}{3}$

(C) $\frac{5\pi}{3} < x < \frac{8\pi}{3}$ (D) $0 < x < \pi$

Q2. If $f(x) = \log_e x,\ x \in [1, 3]$. Find the values of 'c' for which function varifies the LMVT.

(A) $2\log_3 e$ (B) $\frac{1}{2}\log_3 e$

(C) $2\log_e 3$ (D) $\frac{1}{2}\log_e 3$

Q3. Mean value theorem is not applicable for following f(x) in the given interval.

(A) $f(x) = x^{1/3}, a = -1, b = 1$

(B) $f(x) = \begin{cases} 2 & x = 1 \\ x^2 & 1 < x < 2,\ a = 1, b = 2 \\ 1 & x = 2 \end{cases}$

(C) $f(x) = x\,|x|\ ;\ a = -1,\ b = 1$

(D) $f(x) = 1/x\ ;\ a = 1,\ b = 4$

Q4. A private telephone company serving a small community makes a profit of Rs. 12 per subscriber, if it has 725 subscribers. It decides to reduce the rate by a fixed sum for each subscriber over 725, thereby reducing the profit by 1 paise per subscriber. Thus, there will be profit of Rs. 11.99 on each of the 726 subscribers, Rs. 11.98 on each 727 subscribers etc.

The number of subscribers which will give the company the maximum profit, is -

(A) 961 (B) 962

(C) 963 (D) None of these

Q5. A point on the curve $\frac{x^2}{4} + \frac{y^2}{16} = 1$ where the tangent is equally inclined to the axes is -

(A) $\left(\frac{2}{\sqrt{5}}, \frac{-8}{\sqrt{5}}\right)$ (B) $\left(\frac{2}{\sqrt{5}}, \frac{8}{\sqrt{5}}\right)$

(C) $\left(\frac{-2}{\sqrt{5}}, \frac{8}{\sqrt{5}}\right)$ (D) $\left(\frac{-2}{\sqrt{5}}, \frac{-8}{\sqrt{5}}\right)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

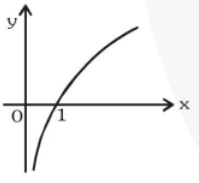
Paragraph # 1 (Ques. 6 to 8)

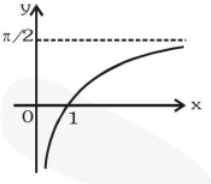
$f:(0,\infty)\to\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ be defined as,

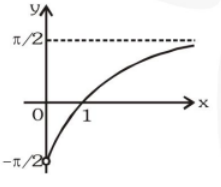
$f(x) = \text{arc tan}(\ell nx)$

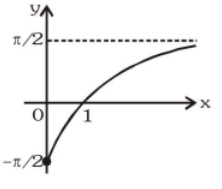
Q6. The above function can be classified as
(A) injective but not surjective
(B) surjective but not injective
(C) neither injective nor surjective
(D) both injective as well as surjective

Q7. The graph of $y = f(x)$ is best represented as -

(A) 

(B) 

(C) 

(D) 

Q8. If x_1, x_2 and x_3 are the points at which $g(x) = [f(x)]$ is discontinuous where [.] denotes greatest integer function, then $x_1 + x_2 + x_3$ is -
(A) equal to 2
(B) equal to 3
(C) greater than 3
(D) greater than 2 but less than 3

Paragraph # 2 (Ques. 9 to 11)

Two functions are defined as follows
$f(x) = ax^2 + bx + c$, $a, b, c \in R$, $a \neq 0$
$g(x) = dx^2 + ex + f$, $d, e, f \in R$, $d \neq 0$
Real part of complex roots of both the equations $f(x) = 0$ and $g(x) = 0$ are same.
Minimum value of $f(x)$ is same as negative of maximum value of $g(x)$.

Q9. Which statement is correct -
(A) $bd = e^2$ (B) $bd = ea$
(C) $bc = ef$ (D) None of these

Q10. If $y = f(|x|)$ has only one critical point, then-
(A) minimum value of y is same as minimum value of f(x)
(B) minimum value of y is greater than minimum value of f(x)
(C) minimum value of y is smaller than minimum value of f(x)
(D) (A) or (B) is correct

Q11. If $y = |g(|x|)|$ has three critical points, then -
(A) maximum value of y can be evaluated
(B) minimum value of y is same as minimum value of f(x).
(C) minimum value of y can be same as minimum value of $f(|x|)$
(D) (B) and (C) are correct

Paragraph # 3 (Ques. 12 to 13)

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in R$

Q12. For $a = 1$ if $y = f(x)$ is strictly increasing $x \in R$ then the maximum range of values of b is.

(A) $\left(-\infty,\frac{1}{3}\right]$ (B) $\left(\frac{1}{3},\infty\right)$

(C) $\left[\frac{1}{3},\infty\right)$ (D) $(-\infty,\infty)$

Q.13 For b = 1, if y = f(x) is non monotonic then the sum of all the integral values of a ∈ [1, 100] is -

(A) 4950 (B) 5049

(C) 5050 (D) 5047

Paragraph # 4 (Ques. 14 to 16)

Consider the function $f(x) = x^2 f(1) - xf'(2) + f''(3)$ such that $f(0) = 2$.

Q14. The values of f'(1) is -

(A) 0 (B) 1 (C) 2 (D) –1

Q15. Equation of tangent to y = f(x) at x = 3 is -

(A) y = x – 7 (B) $y = \frac{x}{4} - 7$

(C) y = 4x – 7 (D) none of these

Q16. The angle of intersection of y = f(x) and $y = 2e^{2x}$ is -

(A) $\tan^{-1}\left(\frac{3}{4}\right)$ (B) $\tan^{-1}\left(\frac{4}{3}\right)$

(C) 0 (D) $\tan^{-1}\left(\frac{6}{7}\right)$

Paragraph # 4 (Ques. 17 to 19)

Let y = f(x) be a differentiable function which satisfies $f'(x) = f^2(x)$ and $f(0) = -\frac{1}{2}$. The graph of the differentiable function y = g(x) contains the point (0, 2) and has the property that for each number 'P', the line tangent to y = g(x) at (P, g(P)) intersects x-axis at P + 2.

Q17. If the tangent is drawn to the curve y = f(x) at a point where it cross the y-axis then its equation is-

(A) x – 4y = 2 (B) x + 7y = 2

(C) x + 4y + 2 = 0 (D) None of these

Q18. The function y = g(x) is given by-

(A) $\frac{e^{-x/2}}{2}$ (B) $e^{-x/2}$

(C) $2.e^{-x/2}$ (D) $e^{-x/2} + 2$

Q19. The number of point intersection of y = f(x) and y = g(x)

(A) 4 (B) 0

(C) 2 (D) 1

MATCH THE COLUMN

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q20.

Column I	**Column II**
(A) $x - \frac{x^3}{3} \le \tan^{-1}x$	(P) $x \in R$
(B) $2x \tan^{-1}x \ge \log_e(1 + x^2)$	(Q) $x > 4$
(C) $\tan x > x + x^3/3$	(R) $x \ge 0$
(D) $x^3 - 3x^2 - 9x + 20 > 0$	(S) $x \in \left(0, \frac{\pi}{2}\right)$

Q21. For the function $f(x) = x^4(12\ln x - 7)$ match the following :

Column-I	**Column-II**
(A) If (a, b) is the point of inflection then a – b is equal to	(P) 3
(B) If e^t is a point of minima then 12t is equal to	(Q) 1
(C) If graph is concave downward in (d, e) then d + 3e is equal to	(R) 4
(D) If graph is concave upward in (p, ∞) then the least value of p is equal to	(S) 8

Q22. Four points A, B, C and D lie in the order on the parabola $y = ax^2 + bx + c$ and the cooordinates of A, B, D are known A(–2, 3); B(–1, 1); D(2, 7).

Column I	**Column II**
(A) The value of $a + b + c =$	(P) –1
(B) If roots of the equation $ax^2 + bx + c = 0$ are α & β then $\alpha^{19} + \beta^7 =$	(Q) 8
(C) Least value of $(a + 2)x^2 + \frac{(b+2)}{x^2} + c =$	(R) 3
(D) If area of quadrilateral ABCD is greatest and co-ordinates of C are (p, q) then $2p + 4q =$	(S) 7

Q23.

Column-I	**Column-II**
(A) The slope of the curve $2y^2 = ax^2 + b$ at (1, –1) is –1, then	(P) a – b = 2
(B) If (a, b) be the point on the curve $9y^2 = x^3$ where normal to the curve makes equal intercepts with the axes, then	(Q) a – b = 7/2
(C) If the tangent at a point (1, 2) on the curve $y = ax^2 + bx + \frac{7}{2}$ be parallel to the normal at (–2, 2) on the curve $y = x^2 + 6x + 10$, then	(R) a – b = 4/3
(D) If the tangent to the curve xy + ax + by = 0 at (1, 1) is inclined at an angle $\tan^{-1} 2$ with x-axis, then	(S) a – b = 3

Q24.

Column-I	**Column-II**
(A) The angle of intersection of $y^2 = 4x$ and $x^2 = 4y$ is 90° & $\tan^{-1}\left(\frac{m}{n}\right)$ then $\lvert m + n \rvert$ is equal to (m and n are co-prime)	(P) 0
(B) The area of triangle formed by normal at the point (1, 0) on the curve $x = e^{\sin y}$ with axes is	(Q) 1/2
(C) The angle between curve $x^2 y = 1$ and $y = e^{2(1-x)}$	(R) 7
(D) The length of sub-tangent at any point on the curve $y = be^{x/3}$ is equal to	(S) 3

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q25. Find the greatest values of the following functions in the given interval if they exist.
$f(x) = 12x^{4/3} - 6x^{1/3}$, $x \in [-1, 1]$

Q26. If the greatest value of $x^3 - 12x^2 + 45x$ in the interval [0, 7] is 'a', then find the value of '6a'

Q27. If $x + y = 8$ then find the maximum value of xy.

Q28. If the line joining the points (0, 3) and (5, –2) is a tangent to the curve $y = \frac{c}{x+1}$, then find the value of c?

Q29. The normal at the point $P\left(2, \frac{1}{2}\right)$ on the curve $xy = 1$, meets the curve again at Q. If m is the slope of the curve at Q, then find $|m|$.

SUBJECTIVE TYPE

Q30. Verify Rolle's theorem for $f(x) = \sin 2x$ in $[0, \pi]$

Q31. Discuss Rolles theorem for $f(x) = [x]$ in $[-1, 1]$

Q32. Discuss Rolles theorem for $f(x) = x^2(x-3)^2$ in [0, 3]

Q33. If $f(x) = ax^3 + bx^2 + 11x - 6$, $x \in [1, 3]$ satisfied condition of rolles theorem and $f'\left(2 + \frac{1}{\sqrt{3}}\right) = 0$, then find a & b.

Q34. For what value of a, m and b does the function $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \le x \le 2 \end{cases}$ satisfy the hypothesis of the mean value theorem for the interval [0, 2].

Q35. Verify LMVT for $f(x) = \tan^{-1}x$ in [0, 1]

Q36. Discuss LMVT for $f(x) = \frac{1}{x}$, $[-1\ 1]$

Q37. Use LMVT to prove that
$1 + x < e^x < 1 + xe^x \ \forall\ x > 0$

***Q38.** Investigate maxima/minima of
$\int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t}\, dt$

Q39. Show that none of the following function has a local maximum or a local minimum :
(i) $x^3 + x^2 + x + 1$ (ii) e^x
(iii) $\log x$ (iv) $\cos x$, $0 < x < \pi$

Q40. Find all the points of local maxima and local minima of the function
$f(x) = x^3 - 6x^2 + 12x - 8$.

Q41. Find the local maxima or local minima, if any, of the function $f(x) = \sin x + \cos x$, $0 < x < \frac{\pi}{2}$ using the first derivative test.

Q42. Find the points at which the function f given by $f(x) = (x-2)^4(x+1)^3$ has
(i) local maxima
(ii) local minima
(iii) points of inflexion

Q43. Which of the following functions has extreme points ?
(i) e^x
(ii) $2x - 3$
(iii) $|x|$ (iv) x^3

Q44. If the sum of the lengths of the hypotenuse and another side of a right angled triangle is constant, show that the area of the triangle is a maximum when the angle between these sides is $\pi/3$.

Q45. If in a triangle ABC, the side 'c' and the angle 'C' remain constant, while the remaining elements are changed slightly, show that $\frac{da}{\cos A}+\frac{db}{\cos B}=0$.

Q46. A man 1.5 m tall walks away from a lamp post 4.5 m high at the rate of 4 km/hr.
(i) How fast is the farther end of the shadow moving on the pavement ?
(ii) How fast is his shadow lengthening ?

Q47. Find the slope of the tangent to the curve $y = x^3 - x$ at $x = 2$

Q48. Find the point at which the tangent to the curve $y = \sqrt{4x-3} - 1$ has its slope $\frac{2}{3}$.

Q49. Find the equation of all lines having slope 2 and being tangent to the curve $y+\frac{2}{x-3}=0$.

Q50. Find the equation of the tangent to the curve $y=\frac{x-7}{(x-2)(x-3)}$ at the point where it cuts the x-axis.

Q51. Find the points on the curve $9y^2 = x^3$ where normal to the curve makes equal intercepts with the axes.

Q52. The slope of the tangent to the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ at the point $t = 2$ is -

Q53. Prove that all normals to the curve $x = a\cos t + at\sin t$, $y = a\sin t - at\cos t$ are at a distance a from the origin.

Q54. If the curves $\frac{x^2}{a^2}+\frac{y^2}{4}=1$ and $y^3 = 16x$ intersect at right angles, show that $a^2=\frac{4}{3}$.

Q55. Find the length of tangent, subtangent normal and subnormal at the point (2, 4) of the curve $y^2 = 8x$.

Q56. Find the length of tangent, sub-tangent normal and sub-normal at the point of the curve $x = a(\theta + \sin\theta)$, and $y = a(1 - \cos\theta)$.

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. $f(x) = 2x + \cot^{-1}x + \ell n(\sqrt{1+x^2} - x)$ is

(A) increasing in $(-\infty, \infty)$

(B) decreasing in $(-\infty, \infty)$

(C) increasing in $(0, \infty)$ and decreasing in $(-\infty, 0)$

(D) decreasing in $(0, \infty)$ and increasing in $(-\infty, 0)$

***Q2.** The interval in which the function $f(x)=\int_{-1}^{x}(t^2+2t)(t^2-1)dt$ decreases is

(A) $(-\infty, -2)$

(B) $(-2,-1)\cup(0,1)$

(C) $(-1,0)\cup(1,\infty)$

(D) None of these

Q3. The value of k in order that $f(x) = \sin x - \cos x - kx + b$ decreases for all real values x, is given by

(A) $k \le 1$ (B) $k \ge 1$

(C) $k \ge \sqrt{2}$ (D) $k \le \sqrt{2}$

Q4. If $f''(x) > 0, \forall x \in R, f'(3) = 0$ and $g(x) = f(\tan^2 x - 2\tan x + 4), 0 < x < \frac{\pi}{2}$, then g(x) is increasing in -

(A) $\left(0, \frac{\pi}{4}\right)$ (B) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

(C) $\left(0, \frac{\pi}{3}\right)$ (D) None of these

Q5. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in-

(A) $(\pi/4, \pi/2)$ (B) $(-\pi/2, \pi/4)$

(C) $(0, \pi/2)$ (D) $(-\pi/2, \pi/2)$

Q6. Let the function $g : (-\infty, \infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2\tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is -

(A) even and is strictly increasing in $(0, \infty)$

(B) odd and is strictly decreasing in $(-\infty, \infty)$

(C) odd and is strictly increasing in $(-\infty, \infty)$

(D) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$

Q7. The values of k for which the function $f(x) = kx^3 - 9x^2 + 9x + 3$ is increasing on R are

(A) $k \geq 3$ (B) $k < 3$

(C) $k \leq 3$ (D) None of these

Q8. The set of possible values of a for which the function $f(x) = x^3 + (a+2)x^2 + 3ax + 5$ is one-one is-

(A) [1, 4] (B) (1, 4)

(C) $(-\infty, 1)$ (D) $(4, \infty)$

Q9. Let $f''(x) > 0 \forall x \in R$ and $g(x) = f(x^2) - f(x^2 + 3)$, then g (x) is -

(A) increasing in $(0, \infty)$

(B) decreasing in $(0, \infty)$

(C) increasing in R

(D) decreasing [–1, 2]

Q10. Let $f(x) = \begin{cases} xe^{ax} & ; \quad x \leq 0 \\ x + ax^2 - x^3 ; & x > 0, \end{cases}$ where a is positive constant. The interval in which f'(x) is increasing is-

(A) $(-\infty, \infty)$ (B) $\left(-\infty, \frac{-2}{a}\right) \cup \left(\frac{a}{3}, \infty\right)$

(C) $\left(-\frac{2}{a}, \frac{a}{3}\right)$ (D) $(0, \infty)$

Q11. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2 - x)$ and $f''(x) < 0 \forall x \in (0, 2)$.

Then g(x) increases in

(A) $\left(\frac{1}{2}, 2\right)$ (B) $\left(\frac{4}{3}, 2\right)$

(C) (0, 2) (D) $\left(0, \frac{4}{3}\right)$

Q12. Let $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$.

Then in $\left(0, \frac{\pi}{2}\right)$.

(A) f(x) is increasing and g(x) is decreasing

(B) f(x) is decreasing and g(x) is increasing

(C) both f(x) and g(x) are increasing

(D) both f(x) and g(x) are decreasing

***Q13.** The difference between the least and the greatest values of the function $f(x) = \int_0^x (\cos^2 t + \cos t + 2)dt$ in interval $[0, 2\pi]$ is $k\pi$, then k =

(A) 1 (B) 3

(C) 5 (D) None of these

***Q14.** The difference between the greatest and the least value of the function $f(x) = \int_0^x \left(6t^2 - 24\right) dt$ on [1, 3] is -

(A) 14 (B) 10

(C) 4 (D) None of these

Q15. If $0 < x < \frac{\pi}{2}$, then

(A) $\cos x < 1 - \frac{x^2}{2}$ (B) $\cos x > 1 - \frac{x^2}{2}$

(C) $\cos x = 1 - \frac{x^2}{2}$ (D) $\cos x \geq 1 + \frac{x^2}{2}$

Q16. Let $f'(x) < 0$ and $g'(x) > 0$ for all real x, then -

(A) $f(g(x+1)) > f(g(x+5))$

(B) $f(g(x)) < f(g(x+1))$

(C) $g(f(x)) < g(f(x+2))$

(D) $g(f(x)) > g(f(x-2))$

Q17. Let $f(x) = x^{\alpha} \log x$ for $x > 0$ and $f(0) = 0$ follows Rolle's theorem for $[0, 1]$ then α is-

(A) –2 (B) –1

(C) 0 (D) 1/2

Q18. If f(x) is twice differentiable function $f(1) = 1, f(2) = 4, f(3) = 9$

(A) $f''(x) = 2$, for atleast one in $x \in (1, 3)$

(B) $f''(x) = f'(x) = 5$, for some $x \in (2, 3)$

(C) $f''(x) = 3, \forall \in (2, 3)$

(D) $f''(x) = 2$, for some $x \in (1, 2)$

Q19. The positive value of k for which $ke^x - x = 0$ has only one root is :-

(A) $\frac{1}{e}$ (B) 1

(C) e (D) $\log_e 2$

Q20. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots is :-

(A) $\left(0, \frac{1}{e}\right)$ (B) $\left(\frac{1}{e}, 1\right)$

(C) $\left(\frac{1}{e}, \infty\right)$ (D) (0, 1)

Q21. The function $f(x) = x(x+3)e^{-(1/2)x}$ satisfies all the conditions of Rolle's theorem in $[-3, 0]$. The value of c is -

(A) 0 (B) –1

(C) –2 (D) –3

Q22. If the function $f(x) = x^3 - 6x^2 + ax + b$ satisfies Rolle's theorem in the interval $[1,3]$ and $f'\left(\frac{2\sqrt{3}+1}{\sqrt{3}}\right) = 0$, then -

(A) $a = -11$ (B) $a = -6$

(C) $a = 6$ (D) $a = 11$

Q23. If $27a + 9b + 3c + d = 0$, then the equation $4ax^3 + 3bx^2 + 2cx + d = 0$ has atleast one real root lying between -

(A) 0 and 1 (B) 1 and 3

(C) 0 and 3 (D) None of these

Q24. Equation $\sin x + 2 \sin 2x + 3 \sin 3x = \frac{8}{\pi}$ has at least one root in

(A) $\left(\pi, \frac{3\pi}{2}\right)$ (B) $\left(0, \frac{\pi}{2}\right)$

(C) $\left(\frac{\pi}{2}, \pi\right)$ (D) None of these

Q25. From mean value theorem

$f(b) - f(a) = (b-a)f'(x_1)$;

$a < x_1 < b$ if $f(x) = \frac{1}{x}$,

then $x_1 = (a, b > 0)$.

(A) $\sqrt{ab}$ (B) $\frac{a+b}{2}$

(C) $\frac{a+b}{2}$ (D) $\frac{b-a}{b+a}$

Q26. The function $f(x) = (x-3)^2$ satisfies all the conditions of mean value theorem in [3, 4]. A point on $y = (x-3)^2$, where the tangent is parallel to the chord joining (3, 0) and (4, 1) is -

(A) $\left(\frac{7}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{7}{2}, \frac{1}{4}\right)$

(C) (1, 4) (D) (4, 1)

Q27. The value of c in Lagrange's theorem for the function $f(x)=\begin{cases} x\cos\left(\frac{1}{x}\right), & x\neq 0 \\ 0, & x=0 \end{cases}$ in the interval [–1, 1] is -

(A) 0

(B) 1/2

(C) – 1/2

(D) non existent in the interval

***Q28.** If a, b, c be non-zero real numbers such that

$$\int_0^1 (1+\cos^8 x)(ax^2+bx+c)\,dx$$

$$=\int_0^2 (1+\cos^8 x)(ax^2+bx+c)\,dx=0,$$

then the equation $ax^2+bx+c=0$ will have.

(A) one root between 0 and 1 and other root between 1 and 2

(B) both the roots between 0 and 1

(C) both the roots between 1 and 2

(D) None of these

Q29. According to mean value theorem in the interval $x \in [0, 1]$ which of the following does not follow -

(A) $f(x)=\begin{cases} \frac{1}{2}-x, & x<\frac{1}{2} \\ \left(\frac{1}{2}-x\right)^2, & x\geq\frac{1}{2} \end{cases}$

(B) $f(x)=\frac{\sin x}{x}$; $x\neq 0, 1$; $x=0$

(C) $f(x)=x|x|$

(D) $f(x)=|x|$

Q30. Maximum value of $\left(\frac{1}{x}\right)^x$ is

(A) $(e)^e$ (B) $(e)^{1/e}$

(C) $(e)^{-e}$ (D) $\left(\frac{1}{e}\right)^e$

Q31. Let $f(x)=\begin{cases} |x| & \text{for } 0<|x|\leq 2 \\ 1 & \text{for } x=0 \end{cases}$, then at $x = 0$, f has

(A) A local maximum

(B) No local maximum

(C) A local minimum

(D) No extremum

Q32. The minimum value of $[(5+x)(2+x)]/[1+x]$ for non-negative real x is

(A) 12 (B) 1 (C) 9 (D) 8

Q33. One maximum point of $\sin^p x\cos^q x$ is $(p>0, q>0, 0<x<\pi/2)$

(A) $x=\tan^{-1}\sqrt{(p/q)}$

(B) $x=\tan^{-1}\sqrt{(q/p)}$

(C) $x=\tan^{-1}(p/q)$

(D) $x=\tan^{-1}(q/p)$

Q34. Maximum slope of the curve $y=-x^3+3x^2+9x-27$ is

(A) 0 (B) 12 (C) 16 (D) 32

Q35. The least value of the function $f(x)=\tan^{-1}x-\frac{1}{2}\ell nx$ in $\left[\frac{1}{\sqrt{3}},\sqrt{3}\right]$ is -

(A) $\frac{\pi}{6}+\frac{1}{4}\ell n3$ (B) $\frac{\pi}{3}-\frac{1}{4}\ell n3$

(C) $\frac{\pi}{6}-\frac{1}{4}\ell n3$ (D) $\frac{\pi}{3}+\frac{1}{4}\ell n3$

***Q36.** The function

$$f(x)=\int_1^x [2(t-1)(t-2)^3+3(t-1)^2(t-2)^2]\,dt$$

is

(A) Maximum at x = 1

(B) Minimum at $x=\frac{7}{5}$

(C) Neither maximum nor minimum at x = 2

(D) All the above are TRUE

***Q37.** Let $f(x)=\int_0^x \frac{\cos t}{t}dt \ \ (x>0)$; then for $x=(2n+1)\frac{\pi}{2}$, f(x) has

(A) Minima when n = 0, 2, 4, ...

(B) Maxima when n = 0, 2, 4, 6, ...

(C) Neither maxima nor minima when n = – 1, –3, –5, ...

(D) None of these

Q38. If $y = a\log x + bx^2 + x$ has its extreme value at x = 1 and x = 2, then (a, b) is -

(A) $\left(1,\frac{1}{2}\right)$ (B) $\left(\frac{1}{2},2\right)$

(C) $\left(2,-\frac{1}{2}\right)$ (D) $\left(-\frac{2}{3},-\frac{1}{6}\right)$

Q39. The function $f(x)=\sum_{k=1}^{5}(x-k)^2$ assumes minimum value for x given by -

(A) 3 (B) 2

(C) $\frac{5}{2}$ (D) 5

***Q40.** The point of extremum of the function $F(x)=\int_1^x e^{-t^2/2}\left(1-t^2\right)dt$ are -

(A) x = 1, –1 (B) x = –2

(C) x = 0 (D) $x=\frac{1}{2}$

Q41. The minimum value of the function $f(x) = 2|x-2| + 5|x-3|, \ \forall x \in R$ is -

(A) 3 (B) 2

(C) 5 (D) 7

Q42. The greatest height of the graph of the curve y = 6 cosx – 8 sinx above the x -axis is

(A) 4 (B) 8

(C) 10 (D) None of these

Q43. If the function $f(x)=2x^3-9ax^2+12a^2x+1$ where a > 0 attains its maximum and minimum at p and q respectively such that $p^2=q$, then p equals

(A) 3 (B) 1

(C) 2 (D) $\frac{1}{2}$

***Q44.** f(x) and g(x) are two differentiable function in [0, 2] such that $f''(x)-g''(x)=0$, $f'(1)=2$, $g'(1)=4$, $f(2)=3$, $g(2)=9$, then $f(x)-g(x)$ at x = 3/2 is-

(A) 0 (B) 2

(C) 10 (D) –5

Q45. The total number of local maxima and local minima of the function

$$f(x)=\begin{cases}(2+x)^3, & -3<x\le -1\\ x^{2/3} & -1<x<2\end{cases}$$ is :-

(A) 0 (B) 1

(C) 2 (D) 3

Q46. The range of values of k for which the function $f(x)=(2k-3)(x+\tan 2)+(k-1)\left(\sin^4 x+\cos^4 x\right)$ does not posses critical points, is

(A) $\left(-\infty,\frac{4}{3}\right)\cup(2,\infty)$ (B) $\left(\frac{4}{3},2\right)$

(C) $\left(\frac{4}{3},\infty\right)$ (D) $(2,\infty)$

Q47. The range of values of k for which the function $f(x)=\left(k^2-7k+12\right)\cos x+2(k-4)x+\log 2$ does not possess critical points, is

(A) (1, 5)

(B) (1, 5) – {4}

(C) (1, 4)

(D) None of these

***Q48.** If $f(x)=\begin{cases}2x^2+\dfrac{2}{x^2} & 0<|x|\le 2\\ 3 & x>2\end{cases}$, then

(A) $x=1,-1$ are the points of global minima

(B) $x=1,-1$ are the points of local minima

(C) $x=0$ is the points of local minima

(D) No extremum

***Q49.** The set of critical points of the function

$f(x)=x-\ln x+\int_2^x\left(\frac{1}{t}-2-2\cos 4t\right)dt$ is -

(A) $\left\{\frac{\pi}{2},n\pi+\frac{\pi}{6}\right\},n\in I$

(B) $\left\{\frac{\pi}{6},\frac{n\pi}{2}\pm\frac{\pi}{6},n\in N\right\}$

(C) $\{n\pi\}$, $n\in I$

(D) None of these

Q50. Let $f(x)=x^3+x^2-10x,-1\le x<0$

$=\cos x,0\le x<\frac{\pi}{2}$

$=1+\sin x,\frac{\pi}{2}\le x\le\pi$

Then at $x=\frac{\pi}{2}$, f (x) has

(A) A local minimum

(B) A local maximum

(C) No extremum

(D) None of these

Q51. The denominator of a fraction number is greater than 16 of the square of numerator, then least value of the number is

(A) $-1/4$ (B) $-1/8$

(C) $1/12$ (D) $1/16$

Q52. If $A+B=\frac{\pi}{2}$, the maximum value of $\cos A\cos B$ is

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) 1 (D) $\frac{4}{3}$

Q53. 20 is divided into two parts so that product of cube of one quantity and square of the other quantity is maximum. The parts are

(A) 10, 10 (B) 16, 4

(C) 9, 11 (D) 12, 8

Q54. If two sides of a triangle be given, then the area of the triangle will be maximum if the angle between the given sides be

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

Q55. The largest area of a trapezium inscribed in a semi-circle of radius R, if the lower base is the diameter is

(A) $\frac{3\sqrt{3}}{4}R^2$ (B) $\frac{\sqrt{3}}{2}R^2$

(C) $\frac{3\sqrt{3}}{8}R^2$ (D) R^2

Q56. From a fixed point A on the circumference of a circle of radius r, the perpendicular AY is let fall on the tangent at P. The maximum area of the triangle APY is -

(A) r^2 (B) $\frac{3\sqrt{3}}{4}r^2$

(C) $\frac{3\sqrt{3}}{8}r^2$ (D) $\sqrt{3}r^2$

Q57. The image of the interval [–1, 3] under the mapping $f(x) = 4x^3 - 12x$ is

(A) [–2, 0] (B) [–8, 72]

(C) [–8, 0] (D) None of these

Q58. If the roots of the equation $x^3-ax^2+bx-8=0$ are real and positive, then the minimum value of a is (b > 0).

(A) 2 (B) 6

(C) 1 (D) None of these

Rate Measure and Tangent Normal

Q59. A particle moves along the curve $y = x^2 + 2x$. Then the points on the curve where the x and y coordinates of the particle changing at the same rate, are-

(A) $\left(\frac{-3}{4}, \frac{-1}{2}\right)$ (B) $\left(\frac{-1}{2}, \frac{-3}{4}\right)$

(C) $\left(\frac{3}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{3}{4}\right)$

Q60. A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is :

(A) 9/2 (B) 9/7 (C) 7/9 (D) 2/9

Q61. The weight W of a certain stock of fish is given by $W = nw$, where n is the size of stock and w is the average weight of a fish. If n and w change with time t as $n = 2t^2 + 3$ and $w = t^2 - t + 2$, then the rate of change of W with respect to t at $t = 1$ is :-

(A) 1 (B) 5

(C) 8 (D) 13

Q62. A ladder, 5 metre long, standing on a horizontal floor, leans against a vertical wall. If the top of the ladder slides downwards at the rate of 10 cm/sec, then the rate at which the angle between the floor and the ladder is decreasing when the lower end of the ladder is 2 metre from the wall is -

(A) 1/10 radian/sec

(B) 1/20 radian/sec

(C) 20 radian/sec

(D) None of these

Q63. In a ΔABC if sides a and b remain constant such that α is the error in C, then relative error in its area is-

(A) $\alpha \cot C$ (B) $\alpha \sin C$

(C) $\alpha \tan C$ (D) $\alpha \cos C$

Q64. A function $y = f(x)$ has a second order derivative $f''(x) = 6(x - 1)$. If its graph passes through the point (2,1) and at that point the tangent to the graph is $y = 3x - 5$, then the function, is-

(A) $(x - 1)^2$ (B) $(x - 1)^3$

(C) $(x + 1)^3$ (D) $(x + 1)^2$

Q65. The point(s) on the curve $y^3 + 3x^2 = 12y$ where the tangent is vertical, is(are) -

(A) $\left(\pm\frac{4}{\sqrt{3}}, -2\right)$ (B) $\left(\pm\sqrt{\frac{11}{3}}, 1\right)$

(C) (0, 0) (D) $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$

Q66. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$:-

(A) on the left of $x = c$

(B) on the right of $x = c$

(C) at no point

(D) at all points

Q67. The curve $y = ax^3 + bx^2 + cx$ is inclined at 45° to x-axis at (0,0) but it touches x-axis at (1, 0), then the values of a,b, c are given by:-

(A) $a = 1, b = -2, c = 1$

(B) $a = 1, b = 1, c\text{-}- = -2$

(C) $a = -2, b = 1, c = 1$

(D) $a = -1, b = 2, c = 1$

***Q68.** The equation of the tangent to the curve

$$y = \int_{x^2}^{x^3} \frac{dt}{\sqrt{1+t^2}}$$ at $x = 1$ is

(A) $\sqrt{2}\, y + 1 = x$

(B) $\sqrt{3}\, x + 1 = y$

(C) $\sqrt{3}\, x + 1 + \sqrt{3} = y$

(D) None of these

Q69. If the sum of the squares of the intercepts on the axes cut off by tangent to the curve $x^{1/3}+y^{1/3}=a^{1/3},\ a>0$ at $\left(\frac{a}{8},\frac{a}{8}\right)$ is 2, then a =

(A) 1 (B) 2

(C) 4 (D) 8

Q70. The number of tangents to the curve $y^2-2x^3-4y+8=0$ that pass through (1, 2) is -

(A) 0 (B) 2

(C) 1 (D) more than 2

Q71. The number of tangents to the curve $x^{3/2}+y^{3/2}=2a^{3/2}$, which are equally inclined to the axes, is -

(A) 2 (B) 1

(C) 0 (D) 4

Q72. The number of normals to the curve $x^2=4y$ which pass through (1, 2) is

(A) 0 (B) 1

(C) 2 (D) more than 2

Q73. If the normal to the curve $x=a(\cos\theta+\theta\sin\theta),\ y=a(\sin\theta-\theta\cos\theta)$ is at a distance P from the origin,then P =

(A) 1 (B) 2 | a |

(C) | a | (D) none of these

Q74. Tangents are drawn from the origin to the curve y = sin x, then their point of contact lie on the curve.

(A) $\frac{1}{y^2}-\frac{1}{x^2}=1$

(B) $\frac{1}{x^2}+\frac{1}{y^2}=1$

(C) $x^2-y^2=1$

(D) $x^2+y^2=1$

Q75. Tangent to the curve $y=x^2+6$ at a point P(1, 7) touches the circle $x^2+y^2+16x+12y+c=0$ at a point Q. Then the coordinates of Q are -

(A) (–6, –11) (B) (–9, –13)

(C) (–10, –15) (D) (–6, –7)

Q76. The line $\frac{x}{a}+\frac{y}{b}=2$ touches the curve $\left(\frac{x}{a}\right)^n+\left(\frac{y}{b}\right)^n=2$ at the point (a, b) for **:-**

(A) n = 2 only

(B) n = –3 only

(C) n is any real number except 0

(D) None of these

Q77. The co-ordinates of the point on the curve $(x^2+1)(y-3)=x$ where a tangent to the curve has the greatest slope are given by :-

(A) $\left(\sqrt{3},3+\frac{\sqrt{3}}{4}\right)$

(B) $\left(-\sqrt{3},3-\frac{\sqrt{3}}{4}\right)$

(C) (0, 3)

(D) None of these

Q78. At which point on the ellipse $\frac{x^2}{8}+\frac{y^2}{18}=1$ must a tangent be drawn such that the area of the triangle formed by the tangent and the coordinate axes is the smallest :-

(A) (2, 3)

(B) $\left(-\sqrt{6},\frac{3}{\sqrt{2}}\right)$

(C) $\left(\sqrt{6},\frac{3}{\sqrt{2}}\right)$

(D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Let $h(x) = f(x) - \{f(x)\}^2 + \{f(x)\}^3$ for all real values of x. Then

(A) h is increasing if $f(x)$ is increasing

(B) h is increasing if $f'(x) < 0$

(C) h is decreasing if f is decreasing

(D) nothing can be said in general

Q2. If $f''(x) < 0$ in [0, 1] and $\phi(x) = f(x) + f(1-x)$, then

(A) ϕ increases in $\left(\frac{1}{2}, 1\right)$

(B) ϕ decreases in $\left(\frac{1}{2}, 1\right)$

(C) ϕ decreases in $\left(0, \frac{1}{2}\right)$

(D) ϕ increases in $\left(0, \frac{1}{2}\right)$

Q3. Which of the following function are increasing on $(-\infty, \infty)$?

(A) tan x (B) $\frac{x}{x+2}$

(C) $2x^3 - 3x^2 + 6x - 1$ (D) $e^{2x^3 - 3x^2 + 6x - 1}$

Q4. The function $f(x) = 4\sin^3 x - 6\sin^2 x + 12\sin x + 100$ is necessarily

(A) increasing in $\left[0, \frac{\pi}{2}\right]$

(B) decreasing in $\left[0, \frac{\pi}{2}\right]$

(C) increasing in $\left[\frac{\pi}{2}, \pi\right]$

(D) increasing in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Q5. Let $f(x) = x^3 + ax^2 + 6x + \cos 3x$, then f^{-1} will exist if

(A) $a < -3$ (B) $-3 \le a \le 3$

(C) $a > 3$ (D) $-2 < a < 2$

Q6. If $0 < x < \frac{\pi}{2}$ then

(A) cos (sin x) > cos x

(B) cos (sin x) < cos x

(C) cos(sin x) >sin(cos x)

(D) cos(sin x) < sin(cos x)

Q7. Which of the following do not satisfy conditions of Rolle's Theorem ?

(A) $e^x \sin x, \left[0, \frac{\pi}{2}\right]$

(B) $(x+1)^2 (2x-3)^5, \left[-1, \frac{3}{2}\right]$

(C) $\sin|x|, [\pi, 2\pi]$

(D) $\sin\frac{1}{x}, \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Q8. Let $f : R \to R$; $f(x) = 5x^3 - x^2 + 10x + 7$. Then

(A) $f(x) > f(x-1)$ (B) $f(x) \ge 7$

(C) f^{-1} exists (D) $f(5) < f(2)$

***Q9.** For the function $\int_0^x \frac{\sin t}{t} dt$, where x > 0,

(A) maximum occurs at x = nπ, n even

(B) minimum occurs at x = nπ, n odd

(C) maximum occurs at x = nπ, n odd

(D) minimum occurs at x = nπ, n even

***Q10.** The function f(x) =

$$\int_{-1}^{x} t\left(e^{t}-1\right)(t-1)(t-2)^{3}(t-3)^{5}dt$$

has a local minimum at x =

(A) 0 (B) 1

(C) 2 (D) 3

Q11. Let $f(x) = \cos x \sin 2x$ then

(A) Value of local minima of $\underset{x\in(-\pi,\pi)}{f(x)} > -\frac{7}{9}$

(B) Value of local minima of $\underset{x\in(-\pi,\pi)}{f(x)} > -\frac{9}{7}$

(C) Value of local minima of $\underset{x\in[-\pi,\pi]}{f(x)} > -\frac{1}{9}$

(D) Value of local minima of $\underset{x\in[-\pi,\pi]}{f(x)} > -\frac{2}{9}$

Q12. If $f(x) = \begin{cases} 3x^2+12x-1 & -1\le x\le 2 \\ 37-x & 2<x\le 3 \end{cases}$ then

(A) $f(x)$ is increasing on [–1, 2]

(B) $f(x)$ is continuous on [–1, 3]

(C) $f'(2)$ exists

(D) $f(x)$ has the maximum value at x = 2.

***Q13.** $f(x) = \begin{cases} x & , \ 0\le x\le 1 \\ 2-e^{x-1} & , \ 1<x\le 2 \\ x-e & , \ 2<x\le 3 \end{cases}$ and

$g(x) = \int_{0}^{x} f(t)\,dt$, $x \in [0, 3]$ then g(x) has

(A) Local maxima at x = 1 + *l*n2 and local minima at x = e

(B) Local maxima at x = 1 and local minima at x = 2

(C) No local maxima

(D) No local minima

Q14. The set of values of a for which the function

$f(x) = (4a-3)(x+\log 5)$

$+2(a-7)\cot\frac{x}{2}.\sin^{2}\frac{x}{2}$

does not possess critical point is :

(A) $[1,\infty)$ (B) $(2,\infty)$

(C) $\left(-\infty,-\frac{4}{3}\right)$ (D) $(-\infty,-1)$

Q15. A particle is moving in a straight line such that its distance at any time t is

$S = \frac{t^4}{4} - 2t^3 + 4t^2 + 7$, then-

(A) velocity is max at t = $\frac{\left(6-2\sqrt{3}\right)}{3}$

(B) acceleration is min at t = 2

(C) the distance is min at t = 0, 4

(D) None of these

Q16. If OT and ON are perpendiculars dropped from the origin to the tangent and normal to the curve x = a $\sin^3$ t, $y = a\cos^3 t$ at an arbitrary point, then

(A) $4OT^2 + ON^2 = a^2$

(B) the length of the tangent = | y sec t|

(C) the length of the normal = $\left|\frac{y}{\sin t}\right|$

(D) None of these

Q17. If the tangent to the curve $x = 1-3t^2, y = t-3t^3$ at point 't' makes an angle θ with the positive direction of x-axis, then (t > 0)

(A) $\theta = \frac{\pi}{2}$

(B) $\sec\theta = \frac{1+9t^2}{6t}$

(C) $\sec\theta + \tan\theta = 3t$

(D) None of these

Q18. Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'\left(\frac{1}{4}\right) = 0$. Then,

(A) $f''(x)$ vanishes at least twice on [0, 1]

(B) $f'\left(\frac{1}{2}\right) = 0$

(C) $\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right)\sin x\,dx = 0$

(D) $\int_{0}^{1/2} f(t)e^{\sin\pi t}dt = \int_{1/2}^{1} f(1-t)e^{\sin\pi t}dt$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 2 (Ques. 19 to 21)

Consider the functions f (x) and g (x) such that

$f(x) = \frac{x^3}{2} + 1 - x\int_0^x g(t)\,dt$ and $g(x) = x - \int_0^1 f(t)\,dt$.

Both f (x) and g (x) are defined from R → R.

***Q19.** Which one of the following holds good for f(x)?

(A) f (x) is bounded

(B) f (x) has exactly one maxima and one minima

(C) f (x) has a maxima but no minima

(D) f (x) has a minima but no maxima

***Q20.** Minimum distance between the functions f(x) and g(x) is

(A) $\frac{4}{3\sqrt{2}}$ (B) $\frac{7}{6\sqrt{2}}$ (C) $\frac{7}{3\sqrt{2}}$ (D) $\frac{8}{3\sqrt{2}}$

***Q21.** The function g (x)

(A) Is injective but not surjective

(B) Cuts the y-axis at – 3/2

(C) Cuts the y-axis at 3/2

(D) Is neither injective nor surjective

Paragraph # 1 (Ques. 22 to 24)

Let $f(x) = \frac{1}{1+x^2}$. Let m be the slope, a be the x-intercept and b be the y-intercept of a tangent to y = f(x), then-

Q22. Abscissa of the point of contact of the tangent for which m is maximum

(A) $\frac{1}{\sqrt{3}}$ (B) 1

(C) –1 (D) $-\frac{1}{\sqrt{3}}$

Q23. FInd the maximum value of b.

(A) $\frac{9}{8}$ (B) $\frac{3}{8}$

(C) $\frac{1}{8}$ (D) $\frac{5}{8}$

Q24. Find the abscissa of the point of contact of tangent for which $\frac{1}{a}$ is maximum

(A) $\frac{1}{\sqrt{3}}$ (B) 1

(C) –1 (D) $-\frac{1}{\sqrt{3}}$

Paragraph # 3 (Ques. 25 to 27)

Suppose f(x) is a real valued polynomial function of degree 6 satisfying the following condition :

(A) 'f' has minimum value at x = 0 & 2

(B) 'f' has maximum value at x = 1

(C) for all x, $\lim_{x\to 0}\frac{1}{x}\ell n\begin{vmatrix} f(x)/x & 1 & 0 \\ 0 & 1/x & 1 \\ 1 & 0 & 1/x \end{vmatrix} = 2$

Q25. Number of solutions of the equation 8f(x) – 1 = 0 is -

(A) one (B) two

(C) three (D) four

Q26. Range of f(x) is -

(A) $\left[-\frac{32}{15}, \infty\right)$ (B) $\left[-\frac{4}{15}, \infty\right)$

(C) $\left(-\infty, \frac{2}{15}\right]$ (D) none of these

Q27. If the area bounded by y = f(x), x-axis, x = ±1; is $\frac{a}{b}$, where a & b are relatively prime then the value of $\tan^{-1}(a - b)$ is -

(A) π/4 (B) –π/4

(C) π/3 (D) π/6

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R), (S) and (T). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q28. Let $f(x) = (x-1)^m (2-x)^n$; $m, n \in N$ and $m, n > 2$.

Column I	**Column II**
(A) Both $x = 1$ and $x = 2$ are the points of minima if	(P) m is even
(B) $x = 1$ is a point of minima and $x = 2$ is a point of inflection if	(Q) m is odd
(C) $x = 2$ is a point of minima and $x = 1$ is a point of inflection if	(R) n is even
(D) Both $x = 1$ and $x = 2$ are the points of inflection if	(S) n is odd
	(T) m and n both are even

Q29. Let $f(x)$ be a real valued function defined by $f(x) = x^2 - 2|x|$

and $g(x) = \begin{cases} \text{minimum}\{f(t) : -2 < t \le x\}, & x \in (-2, 0) \\ \text{maximum}\{f(t) : 0 \le t \le x\}, & x \in [0, 3) \end{cases}$

Column-I	**Column-II**
(A) $g(x)$ is not continuous at x equal to	(P) -2
(B) $g(x)$ is not derivable at x equal to	(Q) 0
(C) Number of points of local extremum of $g(x)$ is equal to	(R) -1
(D) Global minimum value of $g(x)$ is equal to	(S) 2
	(T) 3

Q30.

Column-I	**Column-II**
(A) If the curves $y = 1 - \cos x$, $x \in (-\pi, \pi)$ and $y = \frac{\sqrt{3}}{2}\lvert x \rvert + a$ touch each other then the number of the possible values of a is equal to	(P) 0
(B) If two curves $y^2 = 4a(x - b_1)$ and $x^2 = 4a(y - b_2)$, where a is a positive constant number and b_1 and b_2 are variables touch each other then their point of contact lies on $xy = ka^2$, where k is equal to	(Q) 2 (R) 3
(C) The point on the parabola $y^2 = 4x$, which is nearest to the circle $x^2 + (y-12)^2 = 1$ has the ordinate equal to	(S) 4
(D) The ordinate of the point(s) on the curve: $y^3 + 3x^2 = 12y$ where the tangent is parallel to y-axis is/are	(T) -2

Q31. **Column-I** **Column-II**

(A) Let $f''(x) > 0 \forall x \in R$ and $g(x) = f(4 - x) + f(2 + x)$ then $g(x)$ increases if x belongs to the interval — (P) $(-\infty, -2)$

(B) The equation $x^3 - 3x + a = 0$ will have exactly one real root if a belongs to the interval — (R) $[0, \infty)$

(C) If $f(x) = \cos x + a^2x + b$ is an increasing function for all values of x, then a belongs to the interval — (S) $(1, \infty)$

(D) If $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$ is increasing for all values of x, then a belongs to the interval — (T) $(2, \infty)$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q32. The value of 'a' for which $f(x) = x^3 + 3(a - 7)x^2 + 3(a^2 - 9)x - 1$ have a positive point of maximum lies in the interval $(a_1, a_2) \cup (a_3, a_4)$. Find the value of $a_2 + 11a_3 + 70a_4$.

Q33. If p(x) be a polynomial of degree 3 satisfying $p(-1) = 10$, $p(1) = -6$ and p(x) has maximum at $x = -1$ and $p'(x)$ has minima at $x = 1$. If distance between the point of local maximum and local minimum of the curve is p, then find the value of p.

Q34. If f(x) is a twice differentiable function such that $f(a) = 0$, $f(b) = 2$, $f(c) = -1$, $f(d) = 2$, $f(e) = 0$ where $a < b < c < d < e$ then the minimum number of zeros of $g(x) = (f'(x))^2 + f''(x) f(x)$ in the interval [a, e] is ?

Q35. The curve $y = ax^3 + bx^2 + cx + 5$, touches the x-axis at P(–2, 0) and cuts the y axis at a point Q, where its gradient is 3. Find the value of $(2a - 4b + c)$.

SUBJECTIVE TYPE

Q36. Let $f(x) = \begin{cases} xe^{ax}, & x \le 0 \\ x + ax^2 - x^3, & x > 0 \end{cases}$, where a is positive constant. Find the interval in which f'(x) is increasing.

Q37. Show that there is no real number k for which the equation $x^3 - 3x + k = 0$ has two distinct roots in (0, 1)

Q38. Find the points of maxima and minima $f(x) = \dfrac{5}{3x^4 - 4x^3 - 36x^2 + 28}$.

Q39. Discuss the extrema of $f(x) = \dfrac{x}{1 + x \tan x}$, $x \in \left(0, \dfrac{\pi}{2}\right)$.

Q40. Find the points of maxima and minima of the function $f(x) = \sin 2x + 2 \cos x$, $x \in [0, 2\pi]$. Also comment about the nature of end points.

Q41. For a given curved surface of a right circular cone when the volume is maximum, prove that the semi vertical angle is $\sin^{-1}\dfrac{1}{\sqrt{3}}$.

Q42. Prove that among all triangles with a given perimeter, the equilateral triangle has the maximum area.

***Q43.** Let α, β be real numbers with $0 \le \alpha \le \beta$ and $f(x) = x^2 - (\alpha + \beta)x + \alpha\beta$ such that $\int_{-1}^{1} f(x)\,dx = 1$.

Find the maximum value of $\int_{0}^{\alpha} f(x)\,dx$.

Q44. Find the shortest distance of the point (0, c) from the parabola $y = x^2$, where $0 \le c \le 5$

***Q45.** A cubic f(x) vanishes at $x = -2$ & has relative minimum/maximum at $x = -1$ and $x = 1/3$.

If $\int_{-1}^{1} f(x)\,dx = \frac{14}{3}$, find the cubic $f(x)$.

Q46. Of all the lines tangent to the graph of the curve $y = \frac{6}{x^2+3}$, find the equations of the tangent lines of minimum and maximum slope.

Q47. If $|f(x_1) - f(x_2)| \le (x_1 - x_2)^2$, for all $x_1, x_2 \in R$. Find the equation of tangent to the curve $y = f(x)$ at the point (1, 2).

Q48. Identify which is greater :

$\frac{1+e^{2/3}}{e^{1/3}}$ or $\frac{1+\pi^{2/3}}{\pi^{1/3}}$.

Q49. A and B are points of the parabola $y = x^2$. The tangents at A and B meet at C. The median of the triangle ABC from C has length 'm' units. Find the area of the triangle in terms of 'm'.

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
A	A	A	A	A	C	D	D	B	D
11	12	13	14	15	16	17	18	19	20
B	B	C	A	A	C	D	D	C	D
21	22	23	24	25	26	27	28	29	30
C	D	B	D	B	D	C	B	A	B
31	32	33	34	35	36	37	38	39	40
A	A	D	A	C	A	A	B	C	A
41	42	43	44	45	46	47	48	49	50
A	C	D	B	C	A	C	C	A	B
51	52	53	54	55	56	57	58	59	60
D	A	C	A	A	A	B	C	D	B
61	62	63	64	65	66				
B	A	B	C	C	A				

Ex - 1A

1. A, B, C **2.** C **3.** A, B **4.** B, C **5.** A, B, C, D
6. D **7.** C **8.** C **9.** B **10.** D
11. D **12.** B **13.** B **14.** A **15.** C
16. D **17. A** **18.** C **19.** D
20. (A) → (Q, R, S); (B) → (P, Q, R, S); (C) → (S); (D) → (Q, S)
21. (A) → (S); (B) → (R); (C) → (P); (D) → (Q)
22. (A) → (R); (B) → (P); (C) → (S); (D) → (Q)
23. (A) → (P); (B) → (R); (C) → (Q); (D) → (S)
24. (A) → (R); (B) → (Q); (C) → (P); (D) → (S) **25.** 18 **26.** 420
27. 16 **28.** 4 **29.** 64 **31.** Applicable $c = \frac{\pi}{4}, \frac{3\pi}{4}$
32. Applicable, $c = \sqrt{\frac{4-\pi}{\pi}}$ **33.** $a = 1$ & $b = -6$ **34.** $a = 3, b = 4$ & $m = 1$
36. Not applicable **38.** Minima at $x = -2, 0, 2$ & Maxima at $x = -1, 1$
40. No point of local maxima as well as no point of local minima
41. local maximum at $x = \frac{\pi}{4}$
42. $x = \frac{2}{7}$ is a point of local maximum.; $x = 2$ is a point of local minimum.
$x = -1$ is a point of inflexion.
43. (i) No (ii) No (iii) $x = 0$ is a minimum point. (iv) No
46. (i) 6 km/h; (ii) 2 km/hr **47.** 11 **48.** (3, 2)
49. $y - 2x + 2 = 0$ OR $y - 2x + 10 = 0$ **50.** $20y - x + 7 = 0$
51. (4, 8/3) and (4, –8/3) **52.** $\frac{6}{7}$ **55.** $L_T = 4\sqrt{2}$; $L_N = 4\sqrt{2}$; $L_{ST} = 4$; $L_{SN} = 4$
56. $L_T = 2a\sin\frac{\theta}{2}$; $L_N = a.2\sin^2\frac{\theta}{2}.\sec\frac{\theta}{2}$; $L_{ST} = a\sin\theta$; $L_{SN} = 2a\sin^2\frac{\theta}{2}\tan\frac{\theta}{2}$

Ex - 2									
1	2	3	4	5	6	7	8	9	10
A	B	C	D	B	C	A	A	B	C
11	12	13	14	15	16	17	18	19	20
D	A	C	A	B	A	D	A	A	A
21	22	23	24	25	26	27	28	29	30
C	D	C	B	A	B	D	A	A	B
31	32	33	34	35	36	37	38	39	40
A	C	A	B	B	D	B	D	A	A
41	42	43	44	45	46	47	48	49	50
B	C	C	D	C	A	B	B	B	B
51	52	53	54	55	56	57	58	59	60
B	A	D	D	A	C	B	B	B	D
61	62	63	64	65	66	67	68	69	70
D	B	A	B	D	A	A	A	C	B
71	72	73	74	75	76	77	78		
B	B	C	A	D	C	C	A		

Ex - 2A

1. A, C **2.** B, D **3.** C, D **4.** A, D **5.** B, D
6. A, C **7.** A, D **8.** A, C **9.** C,D **10.** B, D
11. A,B **12.** A, B, D **13.** A,B **14.** B, C **15.** A, B, C
16. A, B, C **17.** B, C **18.** A, B, C, D **19.** D **20.** C
21. B **22.** D **23.** A **24.** A **25.** D
26. A **27.** B
28. (A) $\rightarrow$ (P,R,T); (B) $\rightarrow$ (P,S); (C)$\rightarrow$ (R,Q); (D) $\rightarrow$ (Q,S)
29. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (Q,S); (C) $\rightarrow$ (Q); (D) $\rightarrow$(R)
30. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (S); (C) $\rightarrow$ (S); (D) $\rightarrow$ (Q)
31. (A) $\rightarrow$ (S,T); (B) $\rightarrow$(P,T) ; (C) $\rightarrow$ (P,Q,S,T); (D) $\rightarrow$(R,S,T)
32. 320 **33.** 1040 **34.** 6 **35.** 5

36. $\left[\frac{-2}{a}, \frac{a}{3}\right]$ **38.** f(x) attains maxima at x = – 2, 3 and minima at x = 0.

39. $x = x_0$ is the only point of maxima for $y = f(x)$, where $(x_0 = \cos x_0)$
40. At the end point x = 0, $f'(0^+) > 0 \Rightarrow x = 0$ is a point of minima.
At the end point $x = 2\pi$, $f'(2\pi^-) > 0 \Rightarrow x = 2\pi$ is a point of maxima.

43. $\frac{\sqrt{6}}{108}$ **44.** $\frac{\sqrt{4c-1}}{2}$ **45.** $f(x) = x^3 + x^2 - x + 2$

46. $y - \frac{3}{2} = \frac{3}{4}(x+1)$; $y - \frac{3}{2} = -\frac{3}{4}(x+1)$

47. y = 2 **48.** $\frac{1+\pi^{2/3}}{\pi^{1/3}}$ is greater **49.** $\frac{m\sqrt{m}}{\sqrt{2}}$

-: Important Notes & Formulas :-